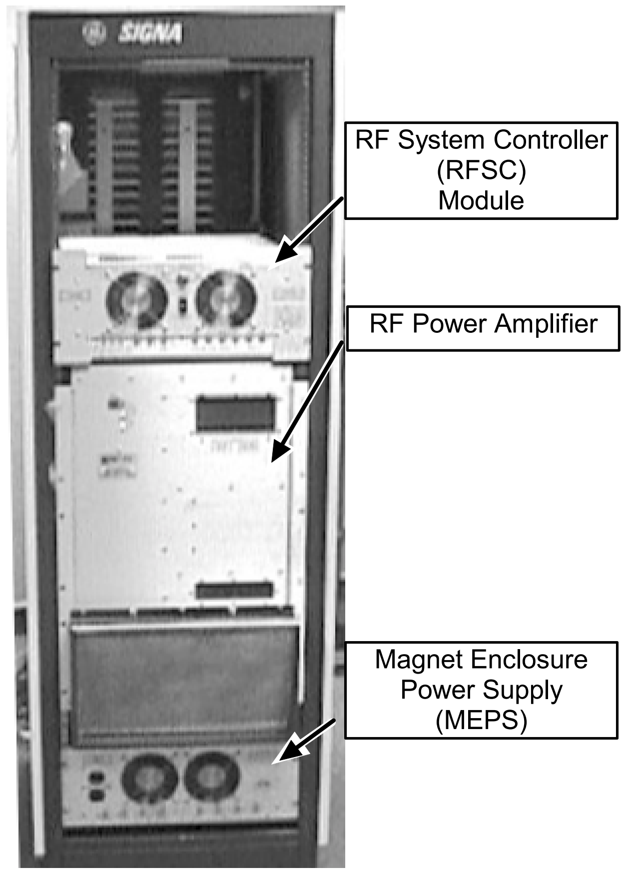
TABLE OF CONTENTS

1- OVERVIEW AND FLOWCHART

This procedure calibrates the RF signal, at the RFSC Analog Processor Board (APB) for the 1.0T and 1.5T RF/PEN I Cabinet, to achieve Body and Head Maximum Power RF Output.



RF/PEN CABINET
ILLUSTRATION 1-1

Note

8x 1.5T System Cabinets ONLY: Some System Cabinets with the 2148300-2 CERD may contain an external Probe Filter (used to correct EFB issues) connected at MR2A11J1 (located inside the rear of the System Cabinet at the I/F Panel). The 2148300-6 CERD contains an internal PROBE Filter. External Probe Filters should not be used with the -6 CERD.

Note

The new RF/Pen 1 Cabinet requires ALL circuit breakers on the rear of the cabinet to be cycled at the same time to prevent an MDS link broke error. The MDS error occurs if you cycle power (RF Amplifier circuit breaker) ONLY on the rear of the RF Amplifier.

The APB controls:

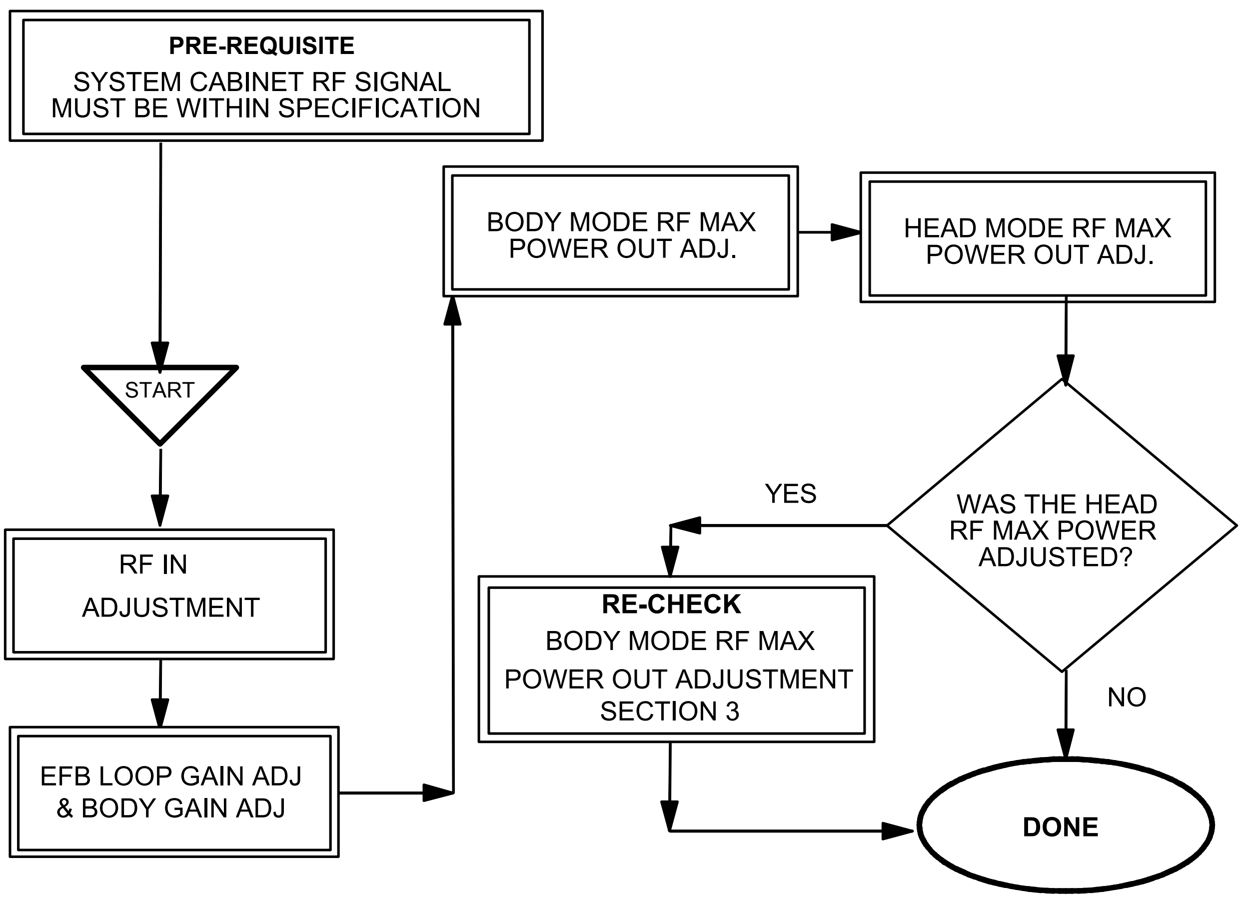
- RF In Balance Pot is set to a specific RF level.
- EFB Loop Gain Adjust Pot is set to a specific DC level and Body Mode Max Power RF Out of the RF Amplifier.

Note

The EFB Loop Gain Adjustment (VCA Control) and the Body Mode RF Power Out Adjustment are interactive.

- Body Mode RF Power Out of the RF Amplifier is set to maximum.
- Head Mode RF Power Out of the RF Amplifier is set to maximum.
- Redundant RF power monitoring for Head, Body (factory set), and Multi-Nuclear Spectroscopy (MNS is not discussed here).

Several test points and pots are accessible on the rear panel of the RFSC, however, this procedure requires opening the RFSC to access the APB test points and pots.



RF MAX POWER OUT

1.5T: BODY MODE, 16.0 kW, 72dBm, (Kit Card # 72)
1.5T: HEAD MODE, 2.0 kW, 63 dBm, (Kit Card # 63)
1.0T: BODY MODE, 7.50 kW, 68.8dBm, (Kit Card # 68.8)
1.0T: HEAD MODE, 750 W, 58.75 dBm, (Kit Card # 58)

CALIBRATION FLOW CHART
ILLUSTRATION 1-2

1-1 Prerequisite — RF Signal (not a required check)

Minimum RF signal levels at the (U)CERD and into the RF/PEN 1 Cabinet. Refer to the following specification Tables and Illustration 1-3 for equipment setup. Setup protocol per Section 2-2.

The System Cabinet 8x ISE CERD Board RF signal must meet specifications per Table 1-1:

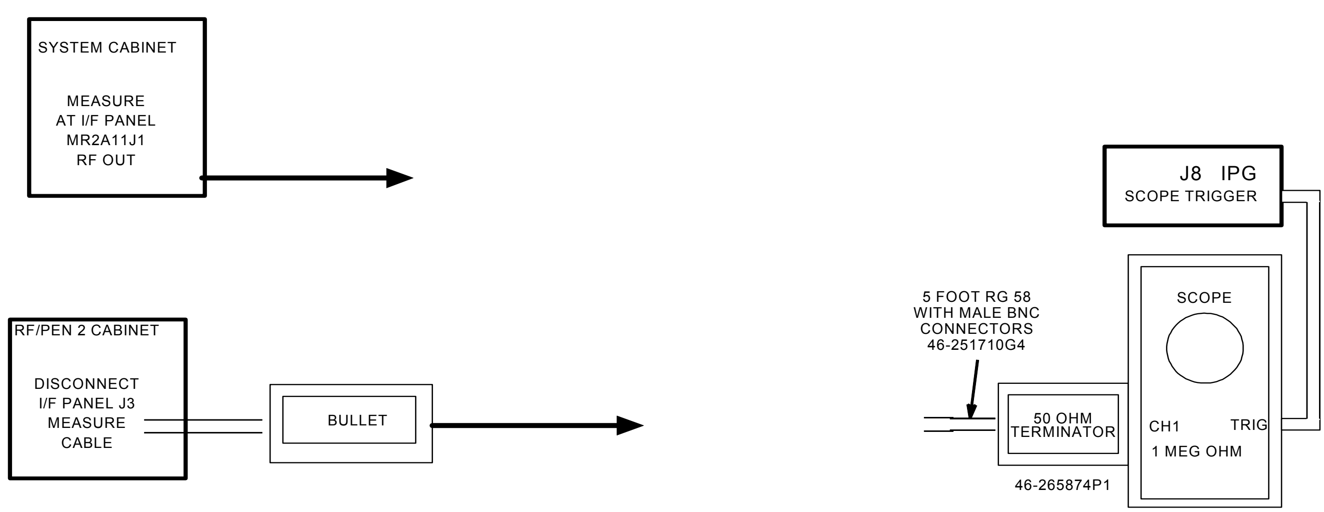
TABLE 1-1
CERD BOARD RF SIGNAL REQUIREMENT

DESCRIPTION	SPECIFICATION — RF POWER MEASUREMENT KIT	ALTERNATE SPECIFICATION
1.5T RF In	≥ 35.7 Minor Divisions (Appendix C, Section C-3)	0.893 to 1.12 VPP 3.0 dBm to 5.0 dBm
1.0T RF In	0.893 VPP to 1.12 VPP (Appendix C, Section C-2)	0.893 to 1.12 VP 3.0 dBm to 5.0 dBm

The minimum RF signal at the RF Cabinet J3 I/F must meet specifications per Table 1-2:

TABLE 1-2
RF CABINET RF SIGNAL REQUIREMENT

DESCRIPTION	SPECIFICATION — RF POWER MEASUREMENT KIT	ALTERNATE SPECIFICATION
1.5T RF In	≥ 15.96 Minor Divisions (Appendix C, Section C-3)	≥ 0.399 VPP - 4.0 dBm minimum
1.0T RF In	≥ 0.399 VPP (Appendix C, Section C-2)	≥ 0.399 VPP - 4.0 dBm minimum



RF SIGNAL MEASUREMENTS
ILLUSTRATION 1-3

2- APB RF ADJUSTMENTS

Section 2 is written as a continuous procedure.

2-1 Tools and Instruments Required

It is recommended that the "Dummy Load and Cables Calibration" procedure is performed before measuring RF Out.

Refer to Table 2-1 for equipment required with the RF Power Measurement Kit or refer to Appendix A (Alternate Equipment Setup).

TABLE 2-1
EQUIPMENT REQUIRED IN ADDITION TO THE RF POWER MEASUREMENT KIT

Item	Description	Part Number
1.	TPS RF Cable Kit DC Block **NOTE:** Not part of the RF Power Measurement Kit	46-301549G1 or 46-301927G1 46-301549P15
2.	100 MHz Scope (equivalent or greater) (oscilloscope probe required): • 468 • 2230 • 2232 • 2465 (300 MHz) • 2465A, 2465B (350 MHz, 400 MHz)	 • 46-183029P61 or 64 • 46-194427P222 • 46-194427P286 or 287 • 46-194427P464 • 46# not supplied
3.	RF Power Measurement Kit **NOTE:** G1kit does contain the 30 dB Dummy Load. • 50 ohm, 200 Watt, 30dB Attenuator Bird Model 8322	46-317724G1 or G2 • 46-255837P10 or • 46-317724P14
4.	Magnetic Shield (for sites with unshielded magnets)	46-317725G10
5.	Pot Tweaker	46-194427P361

2-2 RF In Bal Adjustment at APB

This section will set the RF In Bal. pot to a specific RF level. This signal must be measured into 1 Meg ohm termination at Channel 2 of the oscilloscope.

DANGER!!

PROPERTY DAMAGE! TO PREVENT COIL AND ASSOCIATED SWITCH DAMAGE, REMOVE ALL PHANTOMS AND HARDWARE (I.E., HEAD COIL, SURFACE COIL...) FROM THE MAGNET BORE.

1. Prepare the system to scan as shown in Table 2-2 or refer to Appendix A (Non-Service protocol).

NOTE

If the system is operating with Release 9.X, CNV4 software, or its equivalent then it is no longer necessary to modify the ia_rf1 and ia_rf2 CV values. These will be set automatically when the calmode CV value is set equal to 5.

TABLE 2-2
SCAN PRESCRIPTION - BODY APB CALIBRATION

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

For Non-Service protocol refer to the Appendix A.

A. **[New Pt]**
Id: **geservice <ENTER>**
Name: **apb cals**
Weight (Lb.): **300**
Set Patient Protocols to **Service**.

B. At front enclosure:
Landmark in the Head area—remove any coils.
press **LANDMARK**.
press **MOVE TO SCAN**.

C. In the Patient Position Protocol field:
type **o.41.1** (o=Other, 41.1 =series) to load the body protocol
OR select **other** and select protocol **41** and select series **1**.

D. **[Save Series]**.

E. **[Research Operations]**.
[Setup Params]. Set TG to **50 [Done]**.

F. **[Research Operations]**.
[Display CVs]. Highlight CV Name and enter the following:
CV Name: **calmode<ENTER>**,CV Value: **5<ENTER>** (Dual Logamp Waveform).

NOTE

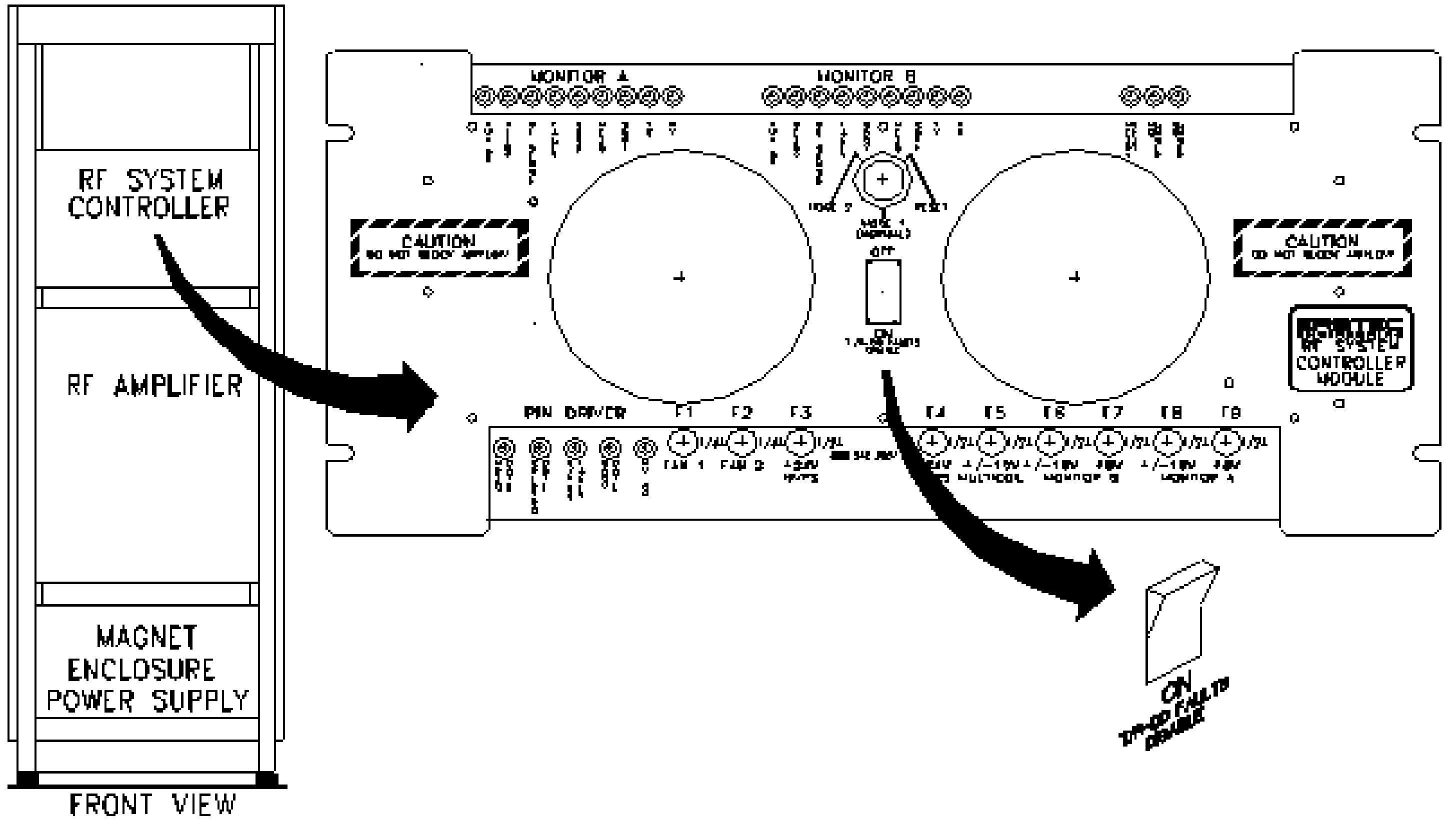
Skip the next two steps if the system has Release 9.X, CNV4, or equivalent software. The software will set the **ia_rf1** and **ia_rf2** CV values automatically once the calmode CV Value is set equal to 5.

CV Name: **ia_rf1<ENTER>**, CV Value: **32766<ENTER>** (sets 90º pulse full scale).
CV Name: **ia_rf2<ENTER>**, CV Value: **0<ENTER>** (turns off 180º pulse).
[Accept].

G. **[Research Operations] [Download]**.

H. **[Prepare to Scan]**.

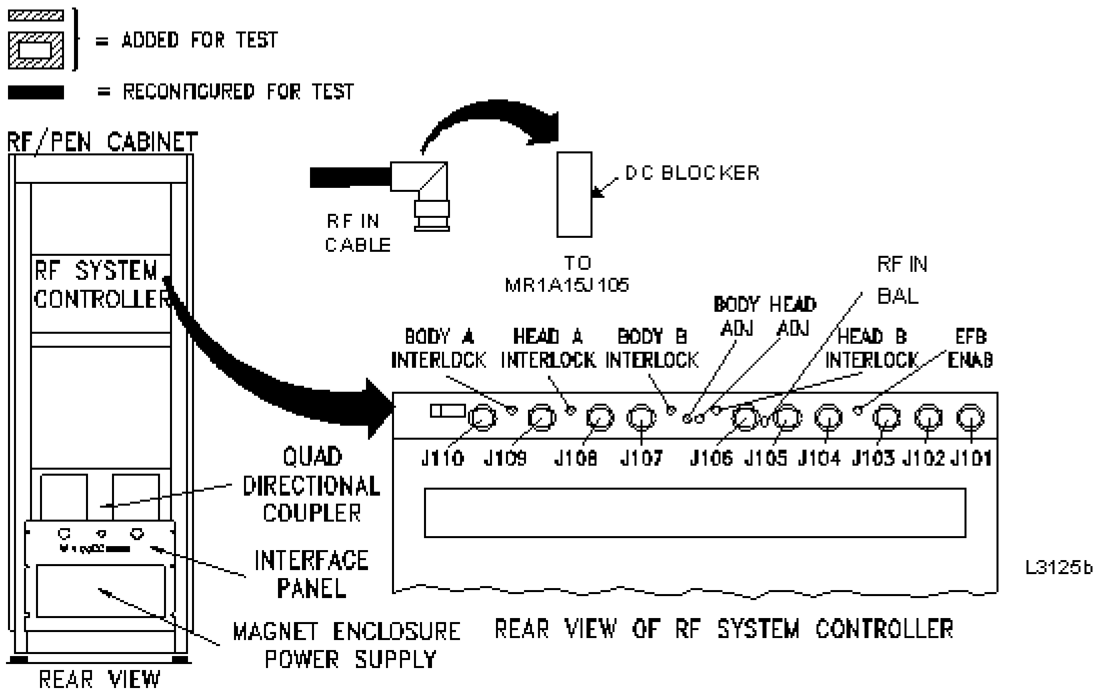
2. See Illustration 2-1. On the RFSC Module (MRIA15A3) front panel, place the T/R-DD Faults rocker switch to the ON (disable fault reporting) position. The RED portion of this switch is a visible indicator the faults reporting is disabled.



T/R- DD FAULTS SWITCH TO ON POSITION
ILLUSTRATION 2-1

3. See Illustration 2-2:

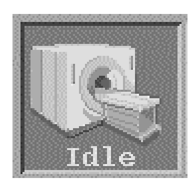
- Disconnect J105 coaxial cable at the rear of the RFSC Module.
- Connect a DC Block in the line at J105.
- Reconnect coaxial cable and DC Block at RFSC Module J105.



DC BLOCK AT J105 RF IN CABLE
ILLUSTRATION 2-2

DANGER!!

PERSONAL INJURY! TO PREVENT POSSIBLE RF BURN WHEN DISCONNECTING HELIAX CABLES FROM “N” HEAD OR “HN” BODY, VERIFY THAT THE SYSTEM IS NOT IN MANUAL PRESCAN OR SCANNING, THE SCAN DESKTOP ICON WILL DISPLAY THE “IDLE,” MESSAGE.

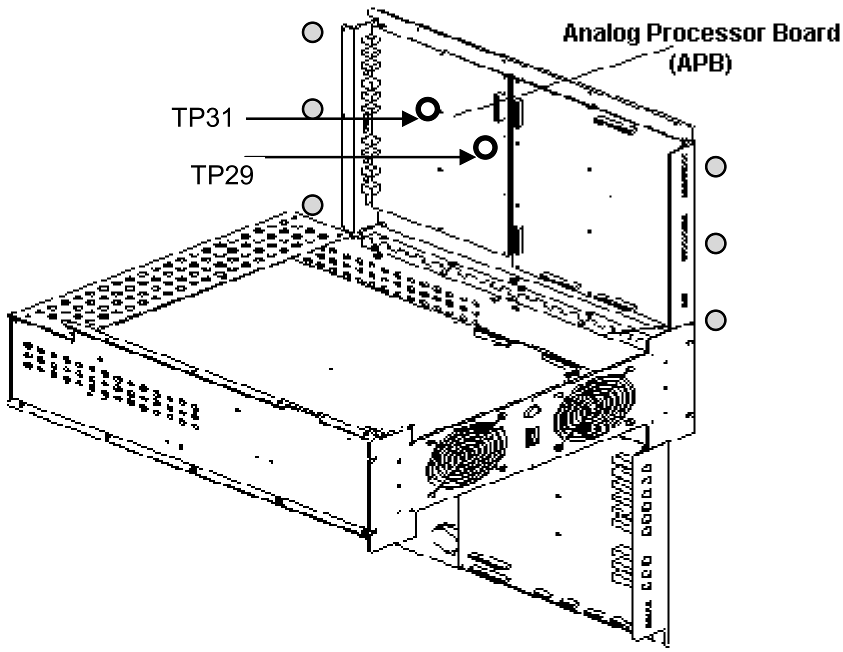


4. Connect the system per RF Power Measurement Kit Card or refer to Appendix A (Alternate Equipment Setup):

- 1.5T — Body Output card 72 (16 kW, 72 dBm), refer to Appendix C, Section C-3.
- 1.0T — Body Output card 68.8 (7.5 kW, 68.8 dBm)), refer to Appendix C, Section C-2.

5. Remove the four 10-32x1/2 panhead screws and four #10 flat washers securing the RFSC Module (MRIA15) to the RF/PEN cabinet front rails, and slide out the module.

6. Remove the 3 screws on the front of the RFCS. Remove the 3 screws on the rear of the RFSC. Loosen the 4 captive screws on the top lid and open. See Illustration 2-3 (gray circles show screw location). Do not open the bottom.



RF SYSTEM CONTROLLER MODULE (RFSC)
ILLUSTRATION 2-3

7. Connect an oscilloscope probe to Channel 2 (1 Meg ohm termination).

8. Connect oscilloscope probe to (Analog Processor Board) TP31 (referenced to ground at TP29 SGND). See Illustration 2-3.

Note

Drape the scope probe cable over the top lid and outside the RFSC Module, instead of over the PIN Switch Driver Board. Move the probe around until the pulse amplitude is maximum. The scope probe may couple with components on the APB and causes a fluctuation in the sync pulse amplitude by as much as 50%.

9. **[Manual Prescan]**.

10. Set TG to 200 (increments of 10 after TG of 150).

11. Adjust the RF IN BAL pot, R270 (Clock-Wise to decrease, Counter-Clock-Wise to increase), until the Channel 2 TP31 measurement is per Table 2-3:

TABLE 2-3
RF IN BAL POT ADJUSTMENT

DESCRIPTION	TP31 SPECIFICATION
1.5T sites and 1.0T sites	2.0 VPP (R270 is located at the rear of the RFSC module) (CW to decrease, CCW to increase).

Note
If the RF Amplifier faults, initially adjust R311, Body Gain (located on the rear of the RFSC Module) several turns CCW to reduce power.

12. Set TG to 0. **[Done]**.

13. Disconnect oscilloscope probe from the APB.

2-3 EFB Loop Gain Adjustment and Body Mode RF Power Output Adjustment

This section contains an **interactive** adjustment. The goal of this section is to successfully achieve the following:

- Channel 1: Achieve 72 dBm, 16 kW Body RF Output (or just slightly less) for 1.5T system or
- Channel 1: Achieve 68.8 dBm, 7.5kW Body RF Output (or just slightly less) for 1.0T system
- Channel 2: The 90 degree pulse portion should not exceed 0.500 VDC.

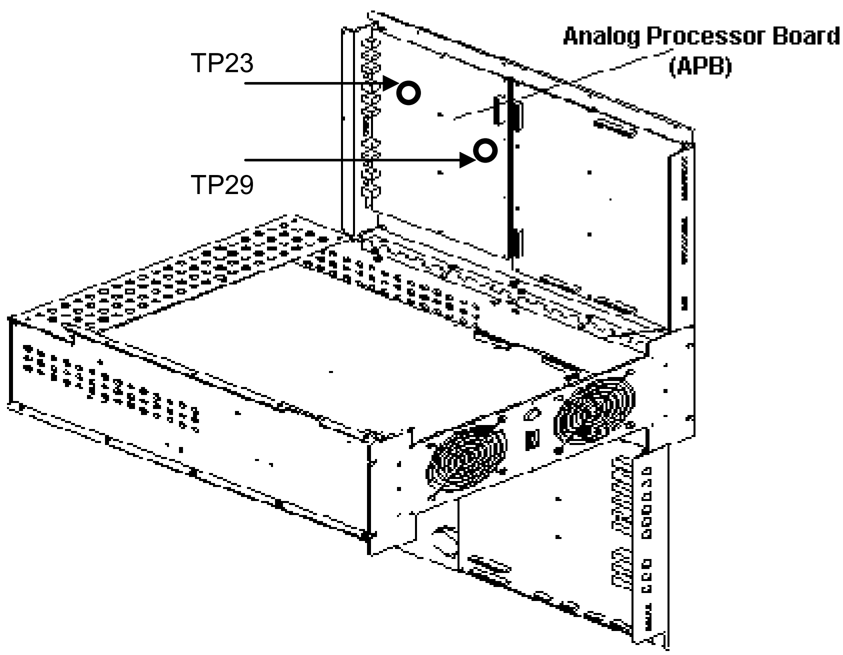
1. Connect scope probe between Channel 2 of the oscilloscope and TP23 (VCA CTRL) (referenced to ground at TP29 SGND) on the APB. See Illustration 2-4.

Note

If R287, 100 Ω potentiometer, is not present, continue with Section 2-4. This pot adjustment was added to minimize a noise issue with the SPT test. It is located on the APB near the EFB Enable green LED.

Note

Drape the scope probe cable over the top lid and outside the RFSC Module.



RF SYSTEM CONTROLLER MODULE (RFSC)
ILLUSTRATION 2-4

2. Oscilloscope set-up for EFB Loop Gain Adjustment as follows:
   - Channel 1: RF Body Power Output signal.
   - Channel 2: 1 Meg Ω input, 0.5 Volts/div, main sweep approximately 1msec/div.
   - Enable the 20 MHz Bandwidth Limit selection for this measurement.
   - Set ground reference for Channel 2 to a known graticule on the oscilloscope display.

3. **[Manual Prescan]**. Initially set TG to 100.

Note

If the RF Amplifier faults due to RF Amplitude: lower the TG, then adjust R311 (Body RF Adjust pot) CCW and R287 (VCA Control pot) CW to decrease the interactive power out. Make small adjustments when tweaking the pots.

4. Monitor Channel 1 (Body RF Out) and Channel 2 (VCA CTRL) at the oscilloscope. Read Table 2-4, Interactive RF Adjustment process:

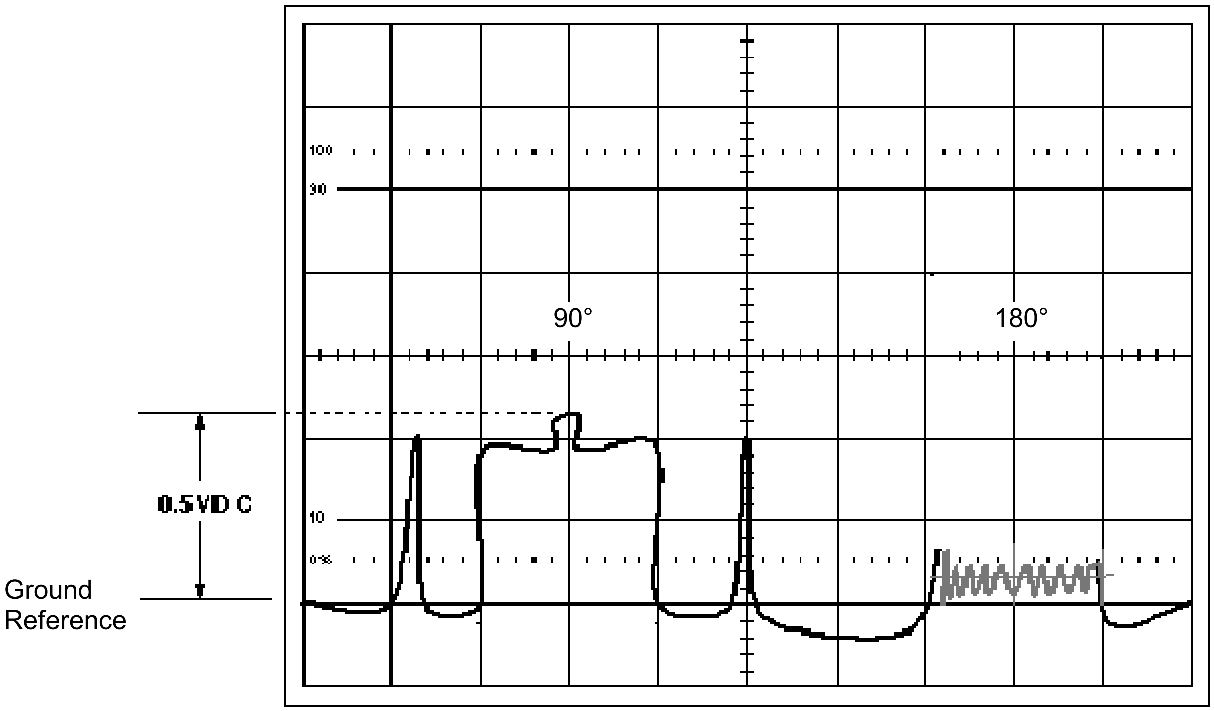
TABLE 2-4
INTERACTIVE RF ADJUSTMENT

<table>
<tr><td colspan="2">CH2: TP23 (with respect to TP29 SGND).
R287: VCA Control pot: CCW to increase, CW to decrease.
NOTE: Do not include any portion of the negative going pulse (below the ground reference) in the measurement. The Illustration is ONLY an example, actual waveforms will vary.</td></tr>
<tr><td>• The 90 degree portion is set to 0.50 VDC as measured with reference to ground.</td><td>• The 180 degree pulse portion (noise) should be set even with the 90 degree low point. This set-point sits at @ 100 mvP or so.
• NOTE: -2 exciter (CERD) without an External PROBE Filter cannot be adjusted. The 180 degree pulse is at @ 1.6 VP above ground reference.</td></tr>
<tr><td colspan="2">CH1: Body RF Output
R311: Body RF Out Adjust pot: CW to increase CCW to decrease.
NOTE: Always disable the 20 MHz Bandwidth Limit selection to view Channel 1.</td></tr>
<tr><td colspan="2">• Achieve slightly less than 72 dBm, 16 kW Body RF Output for 1.5T system OR
• Achieve slightly less than 68.8 dBm, 7.5 kW Body RF Output for 1.0T system</td></tr>
</table>

5. Slowly increase TG (increments of 10) to TG of 200. Adjust the interactive step process above. The final adjusted EFB Loop Gain display should be “similar” to Illustration 2-4.

Note

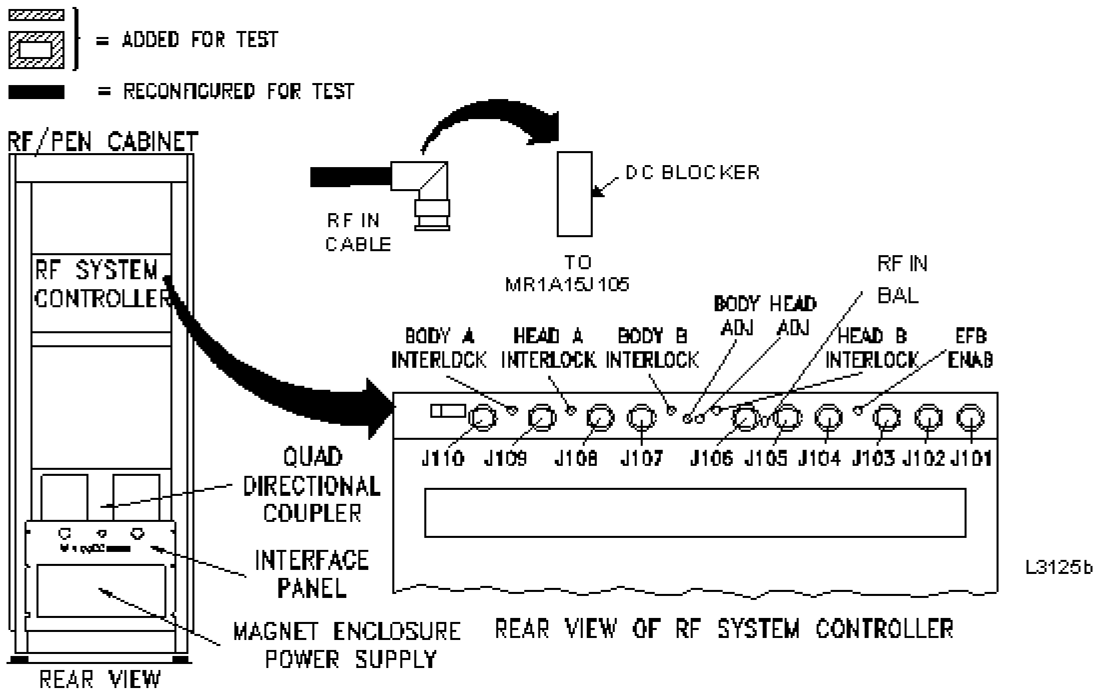
Illustration 2-4 is ONLY for Illustrative purposes. Site waveforms will vary depending on Exciter style. The -2 Exciter without the EXTERNAL PROBE Filter will have a 180 degree pulse at or above 1.5 VP when ia_rf2 is set to zero. This 180 degree pulse cannot be adjusted.



FINAL EFB LOOP GAIN TP23 VCA CTRL WAVEFORM (90 AND 180 PULSE)
ILLUSTRATION 2-4

6. Set TG to 0. **[Done]**.

7. Refer to Illustration 2-5.
   - Disconnect J105 at the RFSC.
   - Remove the DC Block in the J105 line at the rear of the RFSC.
   - Reconnect J105 at the RFSC.



DC BLOCK AT J105 RF IN CABLE
ILLUSTRATION 2-5

8. Remove the oscilloscope probe from the APM Board.

9. Disable the 20 MHz Bandwidth Limit selection.

2-4 Final Body Mode Maximum Power RF Output Adjustment

- 1.5T — Body Mode Max Power RF Output 16 kW, 72 dBm.
- 1.0T — Body Mode Max Power RF Output 7.5 kW, 68.8 dBm.

1. Verify oscilloscope 20 MHz Bandwidth Limit selection is disabled.
2. Verify the DC Block in the J105 line at the rear of the RFSC has been removed. See Illustration 2-5.
3. **[Manual Prescan]**. Initially set TG to 150.
4. View the RF Output power on Channel 1 of the oscilloscope, gradually increase the TG until achieving the Body Max Power RF Out or TG = 200, whichever comes first. Refer to Table 2-5 for process.

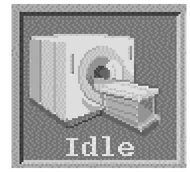
TABLE 2-5
BODY RF OUT ADJ POT PROCESS

SITUATION	ACTION
• TG reaches 200. • Body RF Power is lower than the specified Body Max Power RF Out.	Turn the Body RF Out Adj. pot (R311) CW to increase the RF Power Out.
• TG is less than 200. • Body RF Power Out reaches specified Body Max Power RF Out.	Turn the Body RF Out Adj. pot (R311) CCW to decrease the RF Power Out.

5. Verify with TG set at 200, the Body RF Power Out successfully achieves the specified Body Max Power RF Output.
6. Decrease TG to 0 (zero). **[Done]**.
7. Verify the scan desktop icon is displaying the “Idle” message.
8. Disconnect test equipment at the Body RF Out port.
9. Re-connect the Body RF Output per normal configuration.

2-5 Head Mode Maximum Power RF Output Adjustment

DANGER!!



PERSONAL INJURY! TO PREVENT POSSIBLE RF BURN WHEN DISCONNECTING HELIAX CABLES FROM "N" HEAD OR "HN" BODY, VERIFY THAT THE SYSTEM IS NOT IN MANUAL PRESCAN OR SCANNING, THE SCAN DESKTOP ICON WILL DISPLAY THE "IDLE," MESSAGE.

1. Connect the system per RF Power Measurement Kit Card or refer to Appendix A (Alternate Equipment Setup):
   - 1.5T — Head Output card 63 (2 kW, 63 dBm)), refer to Appendix C, Section C-3.
   - 1.0T — Head Output card 58 (750 W, 58.75 dBm)), refer to Appendix C, Section C-2.

DANGER!!

PROPERTY DAMAGE! TO PREVENT COIL AND ASSOCIATED SWITCH DAMAGE, REMOVE ALL PHANTOMS AND HARDWARE (I.E., HEAD COIL, SURFACE COIL...) FROM THE MAGNET BORE.

2. Prepare the system for scan as shown in Table 2-6 or refer to Appendix A (Non-Service protocol).

NOTE

If the system is operating with Release 9.X, CNV4 software, or its equivalent then it is no longer necessary to modify the ia_rf1 and ia_rf2 CV values. These will be set automatically when the calmode CV value is set equal to 5.

TABLE 2-6
SCAN PRESCRIPTION - HEAD APB CALIBRATION

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.
For Non-Service protocol refer to the Appendix A.

A. **[New Series]**
At Patient Protocols – select **other**.
B. In the protocol field, type **o.41.2<ENTER>** (o=Other, 41.2 =series) to load the head protocol
OR select **[o.41] [Series 2] [Accept]**.
[OK] (if required).
C. **[Save Series]**
D. **[Prepare to Scan]**.
E. **[Research Operations]**.
[Setup Params]. Set TG to **50 [Done]**.
F. **[Research Operations]**.
[Display CVs]. Highlight CV Name and enter the following:
CV name: **calmode<ENTER>**, **5<ENTER>** (Dual Logamp Waveform).

NOTE

Skip the next two steps if the system has Release 9.X, CNV4, or equivalent software. The software will set the **ia_rf1** and **ia_rf2** CV values automatically once the calmode CV Value is set equal to 5.

CV name: **ia_rf1<ENTER>**, **32766<ENTER>** (sets 90º pulse full scale).
CV name: **ia_rf2<ENTER>**, **0<ENTER>** (turns off 180º pulse).
[Accept].
G. **[Research Operations] [Download]**.
H. **[Prepare to Scan]**.

3. **[Manual Prescan]**.

4. Achieve the Head Max Power RF Out at TG = 200 as follows:

- 1.5T — Head Mode Max Power RF Output 2 kW, 63 dBm.
- 1.0T — Head Mode Max Power RF Output 750 W, 58.75 dBm.

5. View the RF Output power on the oscilloscope, gradually increase the TG until achieving the Head Max RF Power Out or TG = 200, whichever comes first. Refer to Table 2-7 for process.

TABLE 2-7
HEAD RF OUT ADJ POT PROCESS

SITUATION	ACTION
• TG reaches 200. • Head RF Power is lower than the specified Head Max Power RF Out.	Turn the Head RF Out Adj. pot (R310) CW to increase the RF Power Out.
• TG is less than 200. • Head RF Power Out reaches specified Head Max Power RF Out.	Turn the Head RF Out Adj. pot (R310) CCW to decrease the RF Power Out.

6. Verify with TG set at 200 the Head RF Power Out successfully achieves the specified Head Max Power RF Output.
7. Decrease TG to 0 (zero). **[Done]**.
8. Verify the scan desktop icon is displaying the “Idle” message.
9. Disconnect test equipment at the Head RF Out port.
10. Re-connect the Head RF Output per normal configuration.

3- RE-CHECK FINAL BODY MODE MAXIMUM POWER RF OUTPUT

1. Repeat section 2-4 (Final Body Mode Maximum Power RF Output Adjustment) if the Head Adj pot was adjusted. Prepare the system to scan per Section 2-2.

4- SYSTEM RESTORATION

1. Verify that the system is not in manual prescan or scanning, the scan desktop icon will display the “Idle” message.

2. On the front panel of the RFSC module place the T/R-DD Faults Fault Rocker Switch to the OFF position (disable fault error reporting set to off). RED is NOT visible.

3. Disconnect test equipment.

4. Replace screws and secure cover of the RFSC.

5. Slide RFSC module into cabinet. Use screws to secure RFSC to front rails.

6. Verify Head and Body heliax cables are connected per normal configuration.

7. Replace all covers and doors to cabinet.

8. Perform one Body Scan to ensure proper operation.

9. Perform one Head Scan to ensure proper operation.

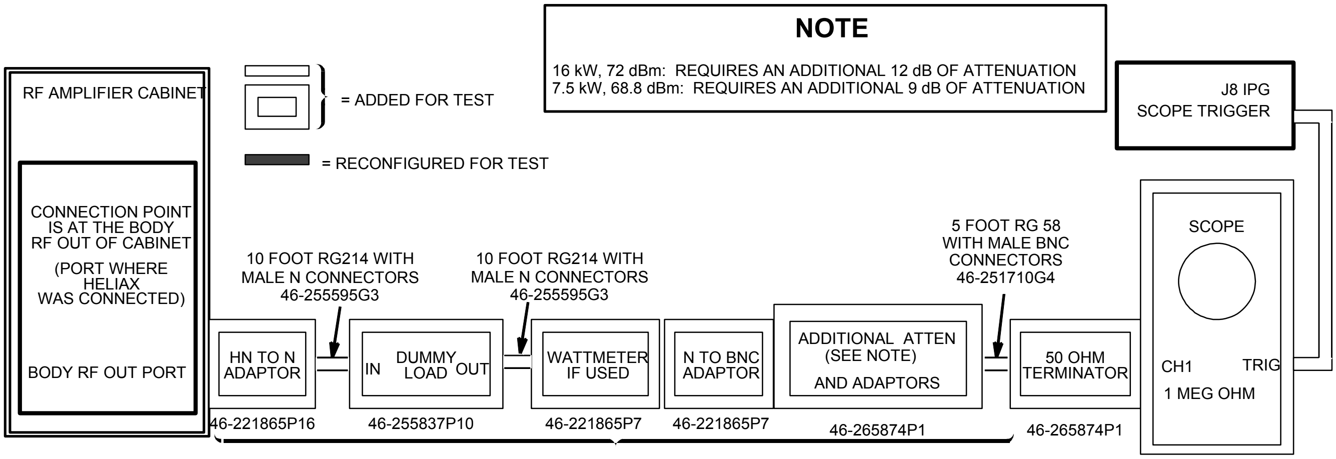
APPENDIX A — ALTERNATE EQUIPMENT SETUPS & MANUAL ENTRY PROTOCOLS

Note

For any RF measurements made using the Alternate Equipment Method a 300 MHz (or greater) oscilloscope is required. This is due to oscilloscope bandwidth limitations.

TABLE A-1
EQUIPMENT REQUIRED IF NOT USING THE RF POWER MEASUREMENT KIT

Item	Description	Part Number
1.	10 dB Rotary Step Attenuator (1 dB step increments) 70 dB Rotary Step Attenuator (10 dB step increments) **NOTE:** May be part of the SST Kit	46-255838P1 46-255838P2
2.	TPS RF Cable Kit BNC male to SMB female test cable	46-301549G1 or 46-301927G1 46-301549P5
3.	RF Test Cable Kit	46-255816G1
4.	100 MHz Scope (equivalent or greater) (1 oscilloscope probe is required): • 468 • 2230 • 2232 • 2465 (300 MHz) • 2465A, 2465B (350 MHz, 400 MHz)	 • 46-183029P61 or P64 • 46-194427P222 • 46-194427P286 or 287 • 46-194427P464 • 46# not supplied
5.	Magnetic Shield (for sites with unshielded magnets)	46-317725G1
6.	Dummy Load: 50 ohm, 200 Watt, 30dB Attenuator Bird Model 8322	46-255837P10 or 46-317724P14
7.	Pot Tweaker	46-194427P361
8.	Wattmeter Kit (optional but not recommended)	46# not supplied



ALTERNATE BODY EQUIPMENT SETUP
ILLUSTRATION A-1

Note

If the wattmeter is not available then bypass it's connection in the circuit using an N-female to N-female connector 46-265875P2.

NOTE

If the system is operating with Release 9.X, CNV4 software, or its equivalent then it is no longer necessary to modify the ia_rf1 and ia_rf2 CV values. These will be set automatically when the calmode CV value is set equal to 5..

NON-SERVICE BODY PROTOCOL

PATIENT REGISTER .	**[New Pt]**
PATIENT INFORMATION	
Patient Id	**geservice**
Patient Name	**body rf**
Weight (Lb)	**300** ¬ *IMPORTANT*
	[Landmark]
Landmark	**[>] [Sternal Notch]**
PATIENT PROTOCOLS	**[Patient Position]**
PATIENT POSITION	
Patient Position	**[>] [Supine]**
Patient Entry	**[>] [Head First]**
Coil	**[...] [Body] [Accept]**
IMAGING PARAMETERS	
Plane	**[>] [Axial]**
Mode	**[>] [2D]**
Pulse Seq	**[...] [Spin Echo]**
	[Accept]
Imaging Options	none (default)
Psd Name	**cal**
Protocol	no entry
SCAN TIMING	
* of Echoes	1 (default)
TE	**[25]**
TR	**[55]**

SCANNING RANGE	
FOV	**[24]**
Slice Thickness	**[5]**
Spacing	**0**
Start	**0**
End	**0**
# Slices	1 (default)
L/R Center	0 (default)
P/A Center	0 (default)
Table Delta	0.00 (default)
ACQUISITION TIMING	
Freq	**[256]**
Phase	**[128]**
NEX	**[2]**
Freq Dir	**[>] [A/P]**
Auto Center Freq	**[>] [Peak]**
(lowest window)	**[Save Series]**

[Research Operations] [Display CVs]

Modify the following: (if 9.X, CNV4, or equivalent only modify **calmode**)

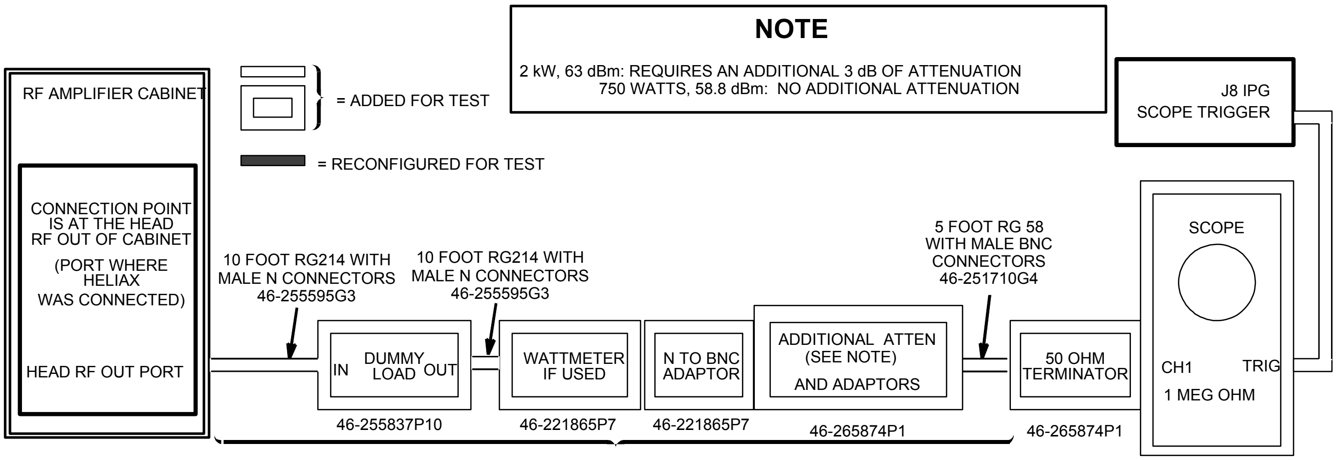
calmode	**5**
ia_rf1	**32766** (90° maximum)
ia_rf2	**0** (180° minimum)
	[Accept]

[Research Operations] [Setup Params]

Set TG **50 [Done]**

[Research Operations] [Download]

[Prepare to Scan]



ALTERNATE HEAD EQUIPMENT SETUP
ILLUSTRATION A-2

Note

If the wattmeter is not available then bypass it's connection in the circuit using an N-female to N-female connector 46-265875P2.

NOTE

If the system is operating with Release 9.X, CNV4 software, or its equivalent then it is no longer necessary to modify the ia_rf1 and ia_rf2 CV values. These will be set automatically when the calmode CV value is set equal to 5.

NON- SERVICE HEAD PROTOCOL

PATIENT REGISTER .	**[New Pt]**
PATIENT INFORMATION	
Patient Id	**geservice**
Patient Name	**head rf**
Weight (Lb)	**300** ¬ *IMPORTANT*
	[Landmark]
Landmark	**[>] [Sternal Notch]**
PATIENT PROTOCOLS	**[Patient Position]**
PATIENT POSITION	
Patient Position	**[>] [Supine]**
Patient Entry	**[>] [Head First]**
Coil	**[...] [Head] [Accept]**
IMAGING PARAMETERS	
Plane	**[>] [Axial]**
Mode	**[>] [2D]**
Pulse Seq	**[...] [Spin Echo]**
	[Accept]
Imaging Options	none (default)
Psd Name	**cal**
Protocol	no entry
SCAN TIMING	
* of Echoes	1 (default)
TE	**[25]**
TR	**[55]**

SCANNING RANGE	
FOV	**[24]**
Slice Thickness	**[5]**
Spacing	**0**
Start	**0**
End	**0**
# Slices	1 (default)
L/R Center	0 (default)
P/A Center	0 (default)
Table Delta	0.00 (default)
ACQUISITION TIMING	
Freq	**[256]**
Phase	**[128]**
NEX	**[2]**
Freq Dir	**[>] [A/P]**
Auto Center Freq	**[>] [Peak]**
(lowest window)	**[Save Series]**

[Research Operations] [Display CVs]

Modify the following: (if 9.X, CNV4, or equivalent only modify **calmode**)

calmode	**5**
ia_rf1	**32766** (90° maximum)
ia_rf2	**0** (180° minimum)
	[Accept]

[Research Operations] [Setup Params]

Set TG **50 [Done]**

[Research Operations] [Download]

[Prepare to Scan]

APPENDIX B — CALIBRATION REQUIREMENTS AFTER HARDWARE REPLACEMENT

TABLE B-1
RE-CALIBRATION REQUIREMENTS AFTER HARDWARE REPLACEMENT

Hardware Replaced	Required Calibration
Exciter Board or CERD	• RF In Bal. Adjustment
External Probe Filter (CERD, -2 exciter module only)	• RF In Bal. Adjustment
RF Amplifier	• EFB Loop Gain Adjustment • Body RF Out Adjustment • Head RF Out Adjustment
APB (Analog Processor Board)	• Entire Procedure
Body Quad Directional Coupler	• Body RF Out Adjustment
Head Quad Directional Coupler	• Head RF Out Adjustment

APPENDIX C — RF POWER MEASUREMENT KIT USE

C-1 RF Formulas and 1.5T Division Conversions

The following RF Calculation Sequences have been provided.

1.5T Division Conversion example from section 1-1:

- 1.5T sites using the 1.5T Scope Calibrator and 100 MHz Oscilloscope equipment:

$$\frac{\text{1.00 VPP (4.0 dBm)}}{\text{Set to 8 Major Div}} = \frac{\text{0.125 VPP each Major Division}}{\text{5 Minor Divisions}} = \text{0.025 VPP each Minor Division}$$

$$\text{3 dBm} = \frac{\text{0.893 VPP}}{\text{0.025 VPP each Minor Division}} = \textbf{35.72 Minor Divisions}$$

VP-P to dBm Calculation:

VP-P **[÷]** 0.632 **[=]** **[LOG]** **[∗]** 20 **[=]** dBm.

dBm to VP-P Calculation:

dBm **[÷]** 20 **[=]** **[INV LOG]** **[∗]** 0.632 **[=]** VP-P.

Example: Using 3.65 dBm, Scope Power reading listed on Kit card

3.65 **[÷]** 20 **[=]** (0.1825) **[INV LOG]** (1.5222991) **[∗]** 0.632 **[=]** (0.9620931) VP-P.

dBm to Watts Calculation:

dBm **[÷]** 10 **[=]** **[INV LOG]** **[=]** **[∗]** 0.001 **[=]** Total Watts.

C-2 RF Power Measurement Kit Use with 1.0T Systems --- Recalculation

The RF Power Measurement Kit contains a 1.5 Tesla Scope Calibrator. This is a 63.86 MHz oscillator that provides a 4 dBm (1.00 VP-P) output into "final" 50 ohms as measured on an oscilloscope. This 1.5T Scope Calibrator is not to be used with 1.0T systems.

Note

1.0T Sites with RF Power Measurement Kit cards that list "Div" (versus VP-P, mVP-P) for the Scope Reading must perform the re-calculation. Additional RF Formulas have been provided.

1.0T Sites with RF Power Measurement Kit cards that have "Div" (scope reading) listed can bypass this Section but must perform the following dBm (scope power) to VP-P calculation:

dBm **[÷]** 20 **[=]** **[INV LOG]** **[∗]** 0.632 **[=]** VP-P.

Example: Using 3.65 dBm, Scope Power reading listed on Kit card

3.65 **[÷]** 20 **[=]** (0.1825) **[INV LOG]** (1.5222991) **[∗]** 0.632 **[=]** (0.9620931) VP-P.

C-3 RF Power Measurement Kit Use with 1.5T Systems

1.5T sites using a 100 MHz oscilloscope must use the RF Power Measurement Scope Calibrator tool to characterize the oscilloscope display. This will compensate for any amplitude error associated with oscilloscope bandwidth limitations or frequency roll-off. The 1.5T Scope Calibrator provides a 63.86 MHz sine wave at +4 dBm/1.00 VPP (as measured on a 300 MHz oscilloscope with “final” 50 ohm termination).

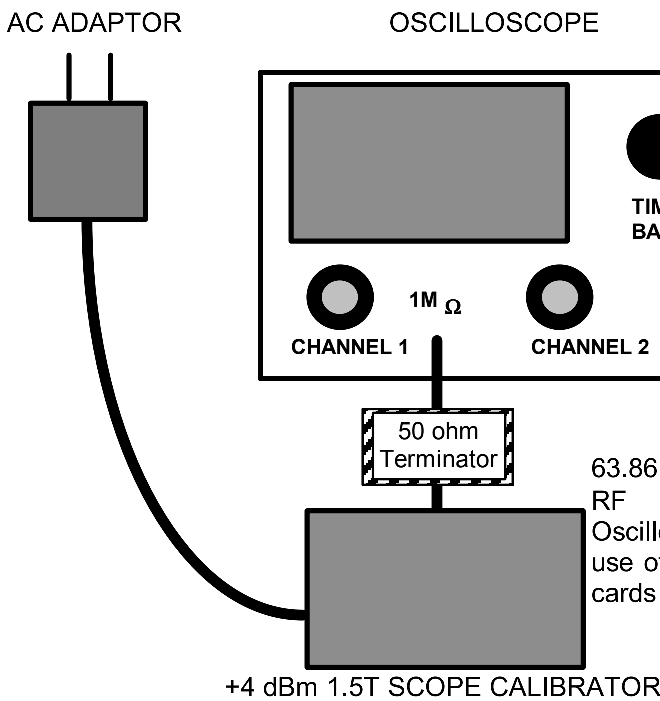
1. Verify the Bandwidth Limit button is not selected on the oscilloscope.

2. Verify Channel 1 is set to 1 M ohm input termination.

Note

When measuring RF signals: Use the 50 ohm terminator supplied with the RF Power Measurement Kit and set Channel 1 oscilloscope termination selection to 1 Meg ohm.

3. Connect the 1.5T Scope Calibrator via the 50 ohm terminator to the oscilloscope input. See Illustration C-1.



Note

63.86 MHz oscillator with 4dBm/1.00 VP-P RF Output into 50 Ω termination. Oscilloscopes less than 300 MHz require the use of this 1.5T Scope Calibrator. All 1.5T cards provide Scope Readings in Divisions.

1.5T SCOPE CALIBRATOR CONNECTIONS
ILLUSTRATION C-1

4. Adjust the channel 1 vertical Volts/div control until amplitude of the waveform is slightly greater than 8 divisions.

5. Adjust the channel 1 vertical Volts/div variable control to achieve exactly 8 divisions.

6. Remove 1.5T Scope Calibrator. Do not remove the 50 ohm terminator attached to the oscilloscope Channel 1 input with 1 Meg ohm oscilloscope termination selected.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 23, 1998	M. Whitlow	Initial Conversion from Toolbook to Word.
1	March 15, 1999	Resa Lambert (KK)	Corrected entire procedure.
2	March 26, 1999	Resa Lambert	Clarified Appendix Section.
3	April 19, 1999	Resa Lambert	Removed all THS730 references.
4	Sept 22, 1999	Resa Lambert	Added TR of “55” to protocols. Removed all THS720 references.
5	Nov 12, 1999	Resa Lambert	Corrected Table 2-4 and Illustration 2-4. Added Note for –2 Exciter without External PROBE Filter which affects 180 degree pulse. CQA997726/SPR# MRIge56705. Changed Non-Service to Manual Entry Protocols in the Appendix section.
6	July 7, 2000	Don Thome’	Corrected head power to 750 W. Added references to 1.0T.
7	Aug. 30, 2001	Don Thome’	Added comments indicating that it is no longer necessary to change ia_rf1 and ia_rf2 with 9.X software or its equivalent.
8	Oct. 24, 2001	Don Thome’	Added comments indicating that it is no longer necessary to change ia_rf1 and ia_rf2 with 9.X, CNV4 software or its equivalent.